

Technology Stack

Date	12 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

1 Frontend / Client-Side

(e.g., React, Angular,

HTML/CSS)
HTML, CSS, Bootstrap, Jinja2
Creates a responsive and user-friendly interface for entering crop parameters and displaying

recommendations.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Handles user requests, processes inputs, and integrates the Machine Learning model.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Joblib (.pkl model)	Stores the crop dataset and the trained machine learning model used for prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render	Hosts the Flask application online, making it accessible through a web browser.
5	Version Control & CI/CD	Git, GitHub	Tracks code changes, enables version control, and stores the project repository.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy	Scikit-learn is used to train the ML model, Pandas handles dataset processing, and NumPy performs numerical computations.